

Online Appendix

Chained Indices Unchained: Structural Transformation and the Welfare Foundations of Income Growth Measurement

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1 Transformation Distortions in Consumption

This appendix summarizes how the benchmark [Herrendorf et al. \(2021\)](#) (HRV 2021) environment changes when there are sector-specific distortions in the transformation of production output into usable consumption goods. The key feature of this extension is that the distortion affects the prices paid by households for final consumption uses, but leaves producer decisions and the investment block unchanged. Hence, the production side, the allocation of capital and labor across sectors, the investment aggregator, and the aggregate technology representation remain the same as in the benchmark model.

1.1 Transformation Technology and Prices

Let P_{jt} denote the producer price of sector $j \in \{g, s\}$, and let P_{jt}^τ denote the price paid by households for usable consumption units. Delivering one unit of usable consumption in sector j requires $e^{\tau_j t}$ units of raw output. Therefore,

$$P_{jt}^\tau = e^{\tau_j t} P_{jt}, \quad j \in \{g, s\}. \quad (1)$$

When $\tau_j = 0$, the benchmark model is recovered. The investment sector purchases raw output at producer prices P_{jt} and is therefore unaffected by the transformation wedge.

Because firms face producer prices, the sectoral value-added producer's problem is identical to the benchmark model. With common factor prices and a common capital share, the capital-labor ratio is equalized across sectors:

$$\frac{K_{gt}}{L_{gt}} = \frac{K_{st}}{L_{st}} = K_t.$$

It follows that relative producer prices remain determined by relative sectoral productivities:

$$\frac{P_{gt}}{P_{st}} = \frac{A_{st}}{A_{gt}}. \quad (2)$$

By contrast, relative expenditure prices satisfy

$$\frac{P_{gt}^\tau}{P_{st}^\tau} = \frac{A_{st}}{A_{gt}} e^{(\tau_g - \tau_s)t}. \quad (3)$$

Thus, the transformation wedge affects structural change only through the prices relevant for household consumption expenditure.

1.2 Investment Block and Aggregate Technology

The investment producer is unchanged. Normalizing the price of investment to one, the unit-cost condition is

$$1 = \frac{[\omega_x P_{gt}^{1-\varepsilon_x} + (1 - \omega_x) P_{st}^{1-\varepsilon_x}]^{1/(1-\varepsilon_x)}}{A_{xt}}. \quad (4)$$

Combining this condition with (2), sectoral producer prices are

$$P_{jt} = \frac{\mathcal{A}_{xt}}{A_{jt}}, \quad j \in \{g, s\},$$

where

$$\mathcal{A}_{xt} = A_{xt} [\omega_x A_{gt}^{\varepsilon_x - 1} + (1 - \omega_x) A_{st}^{\varepsilon_x - 1}]^{1/(\varepsilon_x - 1)}. \quad (5)$$

The expenditure prices faced by households are therefore

$$P_{jt}^\tau = e^{\tau_{jt}} \frac{\mathcal{A}_{xt}}{A_{jt}}.$$

Since $P_{jt} A_{jt} = \mathcal{A}_{xt}$ and the capital-labor ratio is common across sectors, the benchmark aggregation result continues to hold:

$$Y_t = \mathcal{A}_{xt} K_t^\theta. \quad (6)$$

Factor prices are also unchanged relative to the benchmark model:

$$R_t = \theta \mathcal{A}_{xt} K_t^{\theta-1}, \quad W_t = (1 - \theta) \mathcal{A}_{xt} K_t^\theta.$$

Thus, the transformation wedge does not affect production decisions, factor prices, or the aggregate production representation.

1.3 Resource Constraint and Aggregate Balanced Growth Path

The wedge changes the resource constraint for final consumption uses. Sectoral resource feasibility is

$$Y_{jt} = X_{jt} + e^{\tau_{jt}} C_{jt}.$$

Multiplying by producer prices and summing across sectors gives

$$Y_t = X_t + E_t^\tau, \quad E_t^\tau = P_{gt}^\tau C_{gt} + P_{st}^\tau C_{st}. \quad (7)$$

Aggregate output measured at producer prices is therefore allocated between investment and household consumption expenditure measured at expenditure prices.

The aggregate balanced growth path is governed by the benchmark effective productivity $\mathcal{A}_{xt}$. Along an ABGP, the rental rate is constant only if

$$\hat{K}_t = \frac{\hat{\mathcal{A}}_{xt}}{1 - \theta}.$$

Using (6), output growth is

$$\hat{Y}_t = \hat{\mathcal{A}}_{xt} + \theta \hat{K}_t.$$

Hence,

$$\hat{Y}_t = \hat{K}_t = \frac{\hat{\mathcal{A}}_{xt}}{1 - \theta}. \quad (8)$$

The capital accumulation equation implies that X_t/K_t is constant along the ABGP, so $\hat{X}_t = \hat{K}_t$. Combining this result with (7) implies

$$\hat{Y}_t = \hat{X}_t = \hat{K}_t = \hat{E}_t^\tau. \quad (9)$$

Thus, the transformation wedge changes the level and composition of consumption expenditure but not the growth rates of the aggregate objects along the ABGP.

Given an initial capital stock K_0 , the initial level of consumption expenditure that places the economy on the ABGP is

$$E_0^\tau = K_0 \left[\mathcal{A}_{x0} K_0^{\theta-1} - \hat{K} - \delta \right]. \quad (10)$$

This condition is the same resource condition used in the benchmark model, except that consumption expenditure is evaluated at expenditure prices rather than producer prices.

1.4 Measurement

The empirical implementation requires distinguishing producer prices from expenditure prices. Producer prices are the prices relevant for firms' decisions and for the measurement of sectoral value added. Expenditure prices are the prices faced by households when purchasing usable consumption units. Sectoral productivities are therefore recovered from production data using sectoral real value added and factor inputs,

$$A_{jt} = \frac{Y_{jt}}{K_{jt}^\theta L_{jt}^{1-\theta}},$$

while aggregate productivity is recovered from the aggregate Solow residual,

$$\mathcal{A}_{xt} = \frac{Y_t}{K_t^\theta}.$$

Observed relative prices correspond to expenditure prices. We normalize the service wedge to zero, $\tau_s = 0$, and define

$$\zeta \equiv \tau_s - \tau_g = -\tau_g.$$

Using (3), the ratio between the observed relative expenditure price and the model-implied relative producer price satisfies

$$\frac{P_{gt}^\tau / P_{st}^\tau}{A_{st} / A_{gt}} = e^{-\zeta t}. \quad (11)$$

We estimate ζ by fitting an exponential trend to this series. This procedure isolates the component of relative price movements that cannot be accounted for by sectoral productivity differences and attributes it to the transformation wedge. The estimate is

$$\zeta = 0.0137.$$

2 Non-Homothetic CES Preferences

This appendix describes an alternative version of the model in HRV 2021 in which the PIGL demand system is replaced by non-homothetic CES preferences over goods and services. The purpose is to evaluate whether the main quantitative conclusions are robust to a different source of non-homotheticity. The production side of the economy, the investment aggregator, factor prices, and the aggregate technology representation are unchanged relative to the benchmark model. The only modification is on the household side.

2.1 Changes Relative to the Benchmark Model

Let c_{gt} and c_{st} denote consumption of goods and services. Aggregate consumption C_t is implicitly defined by the non-homothetic CES aggregator

$$\omega_c^{1/\sigma} \left(\frac{c_{gt}}{C_t^{\eta_m}} \right)^{\frac{\sigma-1}{\sigma}} + (1 - \omega_c)^{1/\sigma} \left(\frac{c_{st}}{C_t^{\eta_s}} \right)^{\frac{\sigma-1}{\sigma}} = 1. \quad (12)$$

The parameter σ governs substitution between goods and services, ω_c is the goods preference weight, and η_m and η_s control the income elasticity of demand for goods and services, respectively. In the quantitative implementation, we normalize $\eta_m = 1$.

The intratemporal allocation problem gives the expenditure function

$$E(P_{gt}, P_{st}, C_t) = \left[\omega_c P_{gt}^{1-\sigma} C_t^{(1-\sigma)\eta_m} + (1 - \omega_c) P_{st}^{1-\sigma} C_t^{(1-\sigma)\eta_s} \right]^{\frac{1}{1-\sigma}}. \quad (13)$$

Thus, the household's resource cost of attaining aggregate consumption C_t is $E(P_{gt}, P_{st}, C_t)$.

The corresponding expenditure shares are

$$s_{g,t} = \frac{P_{gt} c_{gt}}{E(P_{gt}, P_{st}, C_t)} = \omega_c P_{gt}^{1-\sigma} C_t^{(1-\sigma)\eta_m} E(P_{gt}, P_{st}, C_t)^{\sigma-1}, \quad (14)$$

and

$$s_{s,t} = \frac{P_{st} c_{st}}{E(P_{gt}, P_{st}, C_t)} = (1 - \omega_c) P_{st}^{1-\sigma} C_t^{(1-\sigma)\eta_s} E(P_{gt}, P_{st}, C_t)^{\sigma-1}. \quad (15)$$

The intertemporal problem is then modified as follows. Households have CRRA preferences over the aggregate consumption index,

$$\int_0^\infty e^{-\rho t} \frac{C_t^{1-\psi} - 1}{1-\psi} dt, \quad (16)$$

where ψ is the inverse of the intertemporal elasticity of substitution. The law of motion for capital becomes

$$\dot{K}_t = Y_t - E(P_{gt}, P_{st}, C_t) - \delta K_t. \quad (17)$$

The Euler equation is

$$R_t - \delta - \rho = \psi \frac{\dot{C}_t}{C_t} + \frac{d}{dt} \log \left(\frac{\partial E(P_{gt}, P_{st}, C_t)}{\partial C_t} \right), \quad (18)$$

where R_t is the rental rate of capital, given by

$$R_t = \theta \mathcal{A}_{xt} K_t^{\theta-1},$$

and the last term is the growth rate of the marginal expenditure cost of increasing the non-homothetic consumption index. This marginal expenditure cost is

$$\frac{\partial E(P_{gt}, P_{st}, C_t)}{\partial C_t} = \frac{E(P_{gt}, P_{st}, C_t)}{C_t} [\eta_m s_{gt} + \eta_s s_{st}]. \quad (19)$$

Relative to the benchmark model, the term in (18) plays the same economic role but takes a different form. In the PIGL specification, the marginal expenditure cost can be expressed in closed form because consumption expenditure is measured in service units; its growth is proportional to the growth rate of the service price. Under NHCES preferences, the marginal expenditure cost depends instead on the full expenditure function and on the evolving composition of consumption through the expenditure-share-weighted term $\eta_m s_{gt} + \eta_s s_{st}$.

Solution Algorithm. The main computational implication is that the economy can still be initialized on the aggregate balanced growth path using the same aggregate resource condition as in the benchmark model. Given the exogenous paths for sectoral productivities and prices, we first solve for the initial capital stock K_0 and initial consumption expenditure E_0 using the stationary versions of the capital accumulation equation (17) and the Euler equation (18). These two conditions impose that the economy starts on the aggregate balanced growth path.

Once K_0 and E_0 are determined, the initial aggregate consumption index C_0 is recovered from the NHCES expenditure function,

$$E(P_{g0}, P_{s0}, C_0) = E_0. \quad (20)$$

Unlike in the benchmark PIGL specification, this equation does not generally have a closed-form solution for C_0 , so we solve it numerically.

Given (K_0, E_0, C_0) , the remaining initial allocations, including sectoral consumption demands and expenditure shares, are obtained from the intratemporal demand system. The model is then simulated forward using the dynamic capital accumulation equation (17) and the Euler equation (18).

2.2 Bellman Representation, Indirect Utility, and Expenditure

For welfare calculations, we use the Bellman representation of preferences. Let x denote investment measured in units of the investment good, and let ν denote the marginal

value of investment. Preferences can be written as

$$W(c_g, c_s, x; \nu) = U(c_g, c_s) + \nu x,$$

where

$$U(c_g, c_s) = \frac{C(c_g, c_s)^{1-\psi} - 1}{1 - \psi}.$$

Given income m and prices (P_g, P_s) , the static Bellman problem can be reduced to a one-dimensional choice over the aggregate consumption index:

$$\max_C \left\{ \frac{C^{1-\psi} - 1}{1 - \psi} + \nu [m - E(P_g, P_s, C)] \right\}.$$

Let $C^*(P_g, P_s; \nu)$ denote the solution to this problem. It is characterized by

$$(C^*)^{-\psi} = \nu \frac{\partial E(P_g, P_s, C^*)}{\partial C}.$$

Define

$$\Phi(P_g, P_s; \nu) \equiv \frac{(C^*(P_g, P_s; \nu))^{1-\psi} - 1}{1 - \psi} - \nu E(P_g, P_s, C^*(P_g, P_s; \nu)).$$

Expenditure and Indirect Utility Functions. The indirect utility function associated with the Bellman representation is

$$u(m, P_g, P_s; \nu) = \nu m + \Phi(P_g, P_s; \nu). \quad (21)$$

The corresponding expenditure function is

$$f(w, P_g, P_s; \nu) = \frac{w - \Phi(P_g, P_s; \nu)}{\nu}. \quad (22)$$

The proof follows directly from substituting the budget constraint $m = E(P_g, P_s, C) + x$ into the Bellman representation. Since income enters linearly through νm , the indirect utility function is affine in m , with the price-dependent term summarized by $\Phi(P_g, P_s; \nu)$. Solving (21) for the income level required to attain utility w gives (22).

2.3 Equivalent Variation Measures

Let m_τ denote income in comparison period τ , and let z denote the reference price period. Holding preferences fixed at date t , the equivalent variation measure of income is the

income level $\hat{m}_{t,z,\tau}$ that delivers the same utility as income m_τ at prices $(P_{g,\tau}, P_{s,\tau})$, but evaluated at reference prices $(P_{g,z}, P_{s,z})$.

Using (21), this object is

$$\hat{m}_{t,z,\tau} = m_\tau + \frac{\Phi(P_{g,\tau}, P_{s,\tau}; \nu_t) - \Phi(P_{g,z}, P_{s,z}; \nu_t)}{\nu_t}. \quad (23)$$

This is the non-homothetic CES counterpart of the equivalent variation measure used in the benchmark model. The structure is the same: income is adjusted by a price-correction term. The difference is that, under NHCES preferences, the correction term is computed from the Bellman surplus function Φ , which requires solving a scalar nonlinear equation for $C^*(P_g, P_s; \nu_t)$.

The current-base and reference-base indices are then computed from (23) in the same way as in the benchmark model:

$$\mathcal{P}_{t_1,z} = \log \hat{m}_{t_1,t_1,z} - \log \hat{m}_{t_1,t_1,t_0},$$

and

$$\mathcal{L}_{t_0,z} = \log \hat{m}_{t_0,t_0,z} - \log m_{t_0}.$$

2.4 Calibration

Table 1 summarizes the preference parameters used in the non-homothetic CES specification. The parameter ψ is calibrated to 1.50, following Buera et al. (2011). The parameter η_m is normalized to one. The goods preference weight ω_c is chosen to match the initial goods share in consumption expenditure in 1980. The parameters σ and η_s are jointly identified by the path of consumption expenditure shares. We fix $\sigma = 0.200$ and estimate η_s by minimizing the sum of squared residuals between the model-implied and observed consumption expenditure shares.

As in the benchmark calibration, we also introduce a transformation wedge in consumer prices and calibrate it to match the relative price of goods to services in the data. This wedge affects household consumption prices but leaves the production side and the investment block unchanged, as described in Section 1 of the Online Appendix. All remaining parameters are unchanged from the benchmark calibration.

Figure 1 reports the fit of the non-homothetic CES specification to the goods share in consumption expenditure. The model matches the initial goods share by construction,

Table 1: Calibration of NHCES Preference Parameters

Parameter	Value	Target / Source
ψ	1.500	Calibrated following Buera et al. (2011)
σ	0.200	Fixed (jointly identified with η_s)
ω_c	0.248	Initial goods consumption expenditure share (1980)
η_m	1.000	Normalization
η_s	1.855	Estimated from consumption expenditure shares

through the calibration of ω_c . The subsequent decline is governed by the interaction between relative price movements, the substitution elasticity σ , and the income-effect parameter η_s . Under our calibration, the NHCES specification captures the broad downward trend in the goods expenditure share over the sample period.

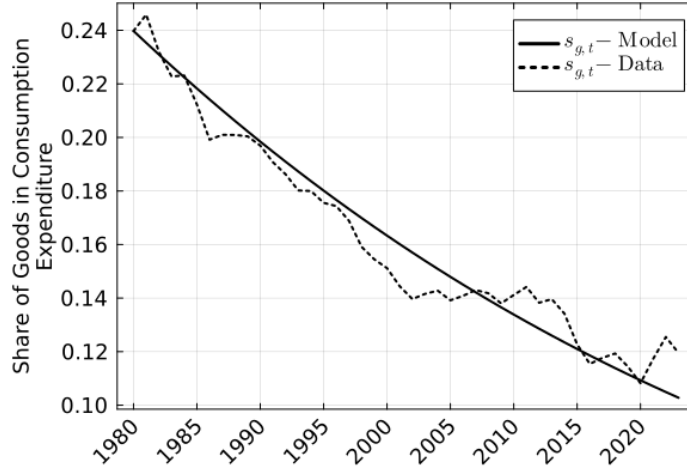


Figure 1: Goods Share in Consumption Expenditure: Model and Data under NHCES Preferences

2.5 Quantitative Results

This subsection reports the quantitative implications of replacing the benchmark PIGL demand system with non-homothetic CES preferences. The production side and the investment block are unchanged, so differences relative to the benchmark model arise only from the household demand system and the associated equivalent variation measures.

Figure 2 is the analogue of the left panel of Figure 5 in the main paper. It reproduces the fixed-base and chained income indices under the non-homothetic CES specification, comparing the chained Divisia index with the current-base and reference-base equivalent

variation measures.

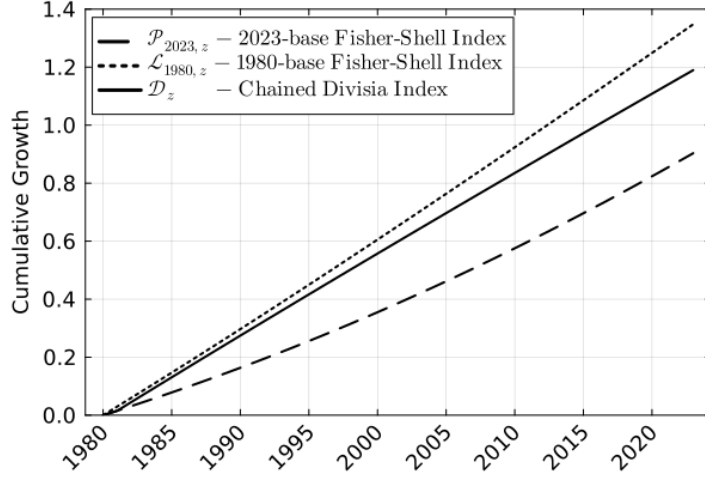
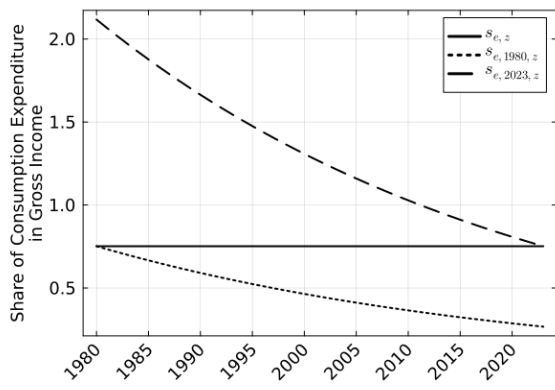


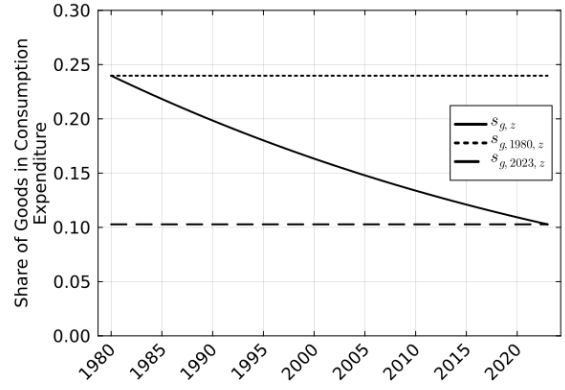
Figure 2: Welfare-Based Real GDP Measures (NHCES Preferences)

As in the benchmark model with PIGL preferences, the 1980-base index lies above the chained Divisia index throughout the sample, while the 2023-base index lies below it. By 2023, the cumulative difference between the 1980-base Fisher–Shell index and the chained Divisia index is 0.152 log points, whereas the cumulative difference between the 2023-base Fisher–Shell index and the chained Divisia index is -0.263 log points.

Figure 3 shows that the equivalent variation measures of income under NHCES preferences share the same properties as in the benchmark PIGL case. Panel (a) reports the consumption expenditure share associated with different bases, while panel (b) reports the corresponding goods consumption share. In both cases, changing the base period systematically changes the measured expenditure and sectoral composition underlying the EV income measures.



(a) Consumption Expenditure Share



(b) Goods Consumption Share

Figure 3: Equivalent Variation Measures for Past- and Current-base Fisher–Shell Indices.

Note: Equilibrium (solid line), 1980-base Fisher–Shell index (dotted), and 2023-base Fisher–Shell index (dashed). For the 1980-base index, $\hat{s}_{e,80,80,z}$ declines from 0.75 in 1980 to 0.27 in 2023; for the 2023-base index, $\hat{s}_{e,2023,2023,z}$ declines from approximately 2.12 in 1980 to 0.75 in 2023. In turn, $\hat{s}_{g,80,80,z}$ is constant at 0.239, while $\hat{s}_{g,2023,2023,z}$ is constant at 0.102.

Figure 4 reports the corresponding growth rates under alternative base periods. This provides the NHCES analogue of the growth-rate comparison reported in Figure 7 in the main text.

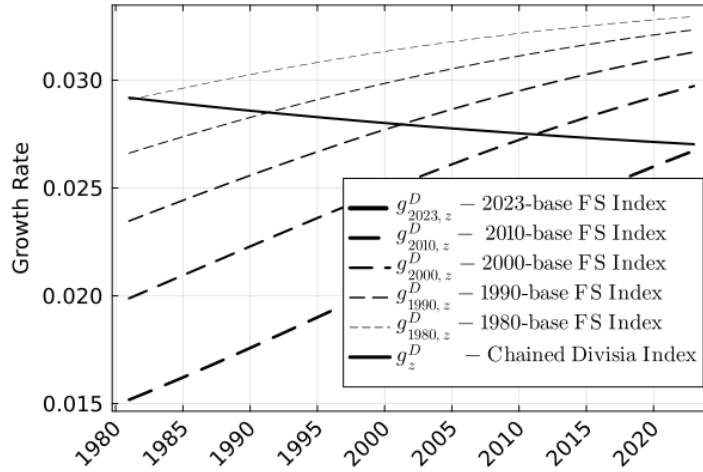


Figure 4: GDP Revisions (NHCES Preferences)

The pattern is the same as in the benchmark specification with PIGL preferences. The reference-base index lies above the chained Divisia index, while the current-base index lies below it. Thus, subsequent revisions systematically lower previous estimates of economic growth, creating the impression that earlier measures had consistently overstated growth.

2.6 Conclusion

The non-homothetic CES extension modifies only the household demand system. The production side, investment aggregator, factor prices, and aggregate technology representation are the same as in the benchmark model. The key difference is that the expenditure function remains available in closed form, while the indirect utility and Bellman surplus functions must be characterized through a scalar nonlinear equation.

Quantitatively, the NHCES specification delivers the same qualitative message as the benchmark PIGL model. Fixed-base equivalent variation measures remain systematically separated from the chained Divisia index, and the direction of the gap depends on the choice of base period. In particular, earlier-base measures imply higher cumulative growth than the chained index, while later-base measures imply lower cumulative growth. Thus, the main welfare-measurement result is not driven by the particular PIGL functional form. Rather, it reflects a more general feature of non-homothetic preferences: changes in relative prices and income effects alter the expenditure and sectoral composition used to evaluate real income growth.

References

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